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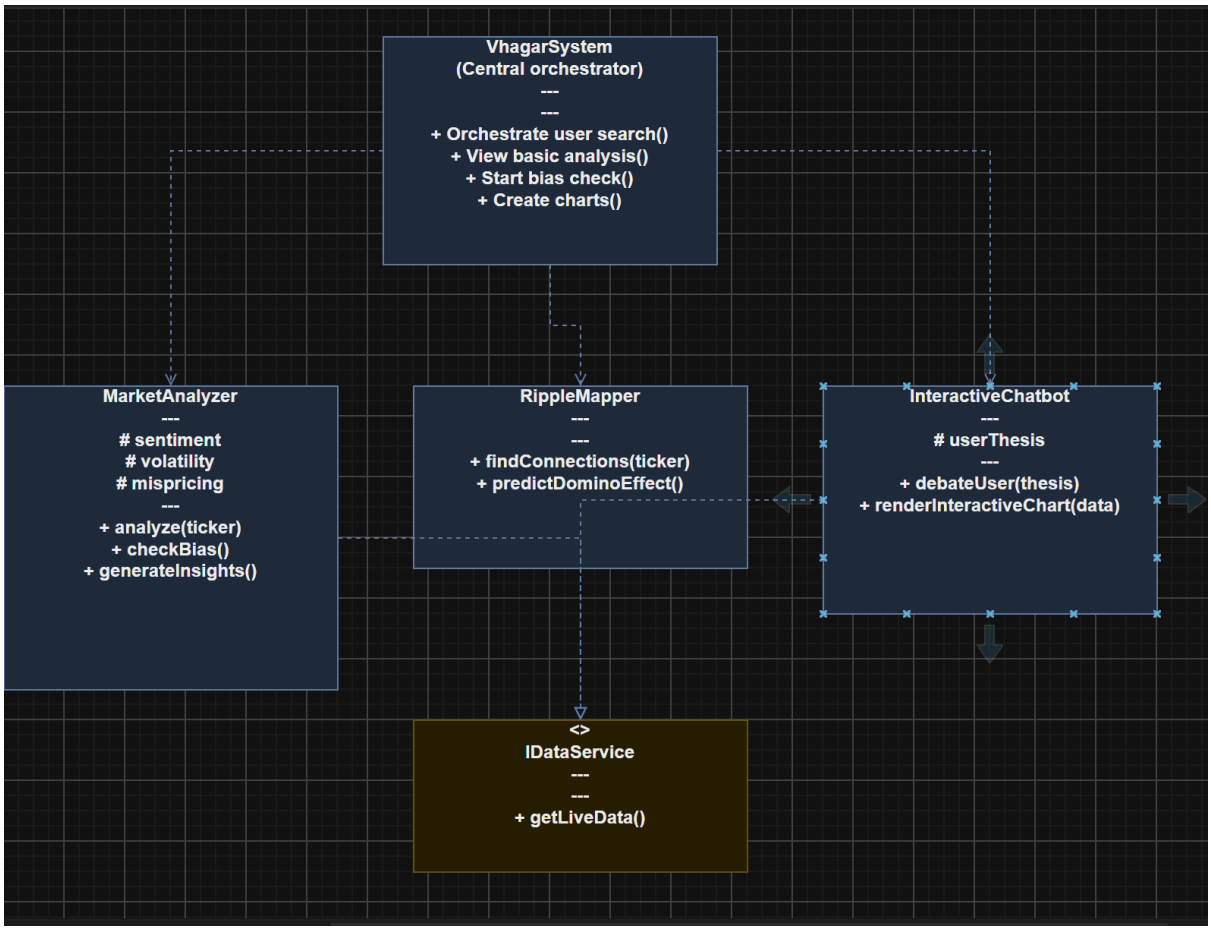
Section : B4_B3_50

Task : Class Diagram of given project

Conventional investment tools fail to synthesize real-time news volatility, supply-chain dependencies, and user cognitive biases into a single actionable workflow.

Vhagar solves this by integrating a sentiment-volatility divergence engine to flag mispricing, a cross-asset surveillance mapper to anticipate supply-chain ripple effects, and an adversarial "Devil's Advocate" chatbot that actively challenges user investment theses with contradictory evidence. To ensure clarity, the system dynamically renders interactive Plotly charts alongside its insights, all unified by a centralized real-time news and search layer that feeds every analytical module—empowering users to make transparent, bias-resistant, and visually informed decisions

CLASS DIAGRAM



1. VhagarSystem (Concrete Class)

- **Role**: Central orchestrator – this is the main entry point that controls the entire workflow.
- **Operations (Methods)**:
    - Orchestrate user search()
    - View basic analysis()
    - Start bias check()
    - Create charts()
- **Relationships**: Depends on (uses) the 3 classes below via dashed arrows.

2. MarketAnalyzer (Concrete Class)

- **Role**: Handles financial market data analysis for a given ticker.
- **Attributes (#)**:
    - sentiment
    - volatility
    - mispricing
- **Operations (Methods)**:
    - analyze(ticker)
    - checkBias()
    - generateInsights()
- **Relationships**: Realizes (implements) the IDataService interface to fetch live data.

3. RippleMapper (Concrete Class)

- **Role**: Maps supply chain connections and predicts knock-on effects.
- **Attributes**: None listed in the diagram.
- **Operations (Methods)**:
    - findConnections(ticker)
    - predictDominoEffect()

- **Relationships**: Realizes (implements) the IDataService interface.

4. InteractiveChatbot (Concrete Class)

- **Role**: Handles conversational interaction, debates user theses, and renders charts.
- **Attributes (#)**:
    - userThesis
- **Operations (Methods)**:
    - debateUser(thesis)
    - renderInteractiveChart(data)
- **Relationships**: Realizes (implements) the IDataService interface.

5. IDataService (Interface — NOT a class)

- **Role**: Acts as a contract to provide real-time/live market data to all analyzers.
- **Operation (Method)**:
    - getLiveData()
- **Relationships**: The three analyzers (MarketAnalyzer, RippleMapper, InteractiveChatbot) realize (implement) this interface, meaning they all must provide their own version of the getLiveData() method.